Enroll. No._____

MARWADI UNIVERSITY

Faculty of Technology

CE-FOT1 (MU), CE, BIOINFO FOT(MU)

B.Tech. SEM:3 **MID-SEM. EXAM: I** **SEPTEMBER-2024**

Subject: - (Probability & Statistics) (01CE0309) **Date:- 17/09/2024**

Total Marks:-30 **Time: - 75 Minutes**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**

Question: 1. [6]

(a) Answer in Short.

1. In Rolling a die what will be the probability of getting an even number?
   a) 0.5 b) 0.1 c) 0.3 d) None of these

2. If $P(B) = 9/13$ & $P(A \cap B) = 4/13$ Find $P(A|B)$?
   a) 4/9 b) 9/4 c) 13 d) None of these

3. If $b_{xy} = -0.1$ and $b_{yx} = -1.1$ then what is value of correlation (r)?
   a) 0.33 b) -0.33 c) 1.1 d) None of these
4. If $P(A) = 0.42$ then what is probability of complement of A ?
   a) 0.58 b) 0.24 c) 1 d) None of these

5. If A and B mutually exclusive events and $P(A) = 3/10$, $P(B) = 2/5$ then find $P(A \cup B)$?
   a) 1 b) 0.7 c) 0.3 d) None of these

6. What is the meaning of No correlation between x and y ?
   a) $r = 1$ b) $r = 0$ c) $r = -1$ d) $r = 1.5$

Question: 2. [12]

(a) Bag I contain 4 white and 6 black balls while Bag II contains 4 white and 3 black balls. One ball is drawn at random from one of the bags, and it is found to be black. Find the probability that it was drawn from i) Bag I ii) Bag II. [06]

(b) Find the regression coefficient b_{xy} and b_{yx} , hence find the correlation coefficient between x *and* y . [06]

X	4	2	3	4	2
Y	2	3	2	4	4

OR

(b) Calculate the coefficient of correlation [06]

X	9	8	7	6	5	4	3	2	1
Y	15	16	14	13	11	12	10	8	9

Question: 3. [12]

a) The competitors in a beauty contest are ranked by three judges in the following order. Use rank correlation coefficient to discuss which pair of judges has the nearest approach to beauty. [08]

1st Judge	**1**	**5**	**4**	**8**	**9**	**6**	**10**	**7**	**3**	**2**
2nd Judge	4	8	7	6	5	9	10	3	2	1
3rd Judge	6	7	8	1	5	10	9	2	3	4

b) A bag contains 30 balls numbered 1 to 30. One ball is drawn at random find the probability that the number will be (i) multiple of 5 or 7 (ii) a multiple of 3 or 7. [04]

OR

a) The number of bacterial cells (y) per unit volume in a culture at different hours (x) is given below. [08]

x	0	1	2	3	4	5	6	7	8	9
y	43	46	82	98	123	167	199	213	245	272

Fit lines of regression of y on x and x on y. Also, estimate the number of bacterial cells after 15 hours.

b) In a certain assembly plant, three machines B1, B2, and B3 make 30%, 45%, and 25% respectively, of the products. It is known from past experience that 2%, 3%, and 2% of the products made by each machine, respectively, are defective. Now, suppose that a finished product is randomly selected. What is the probability that it is defective? [04]

---Best of Luck---

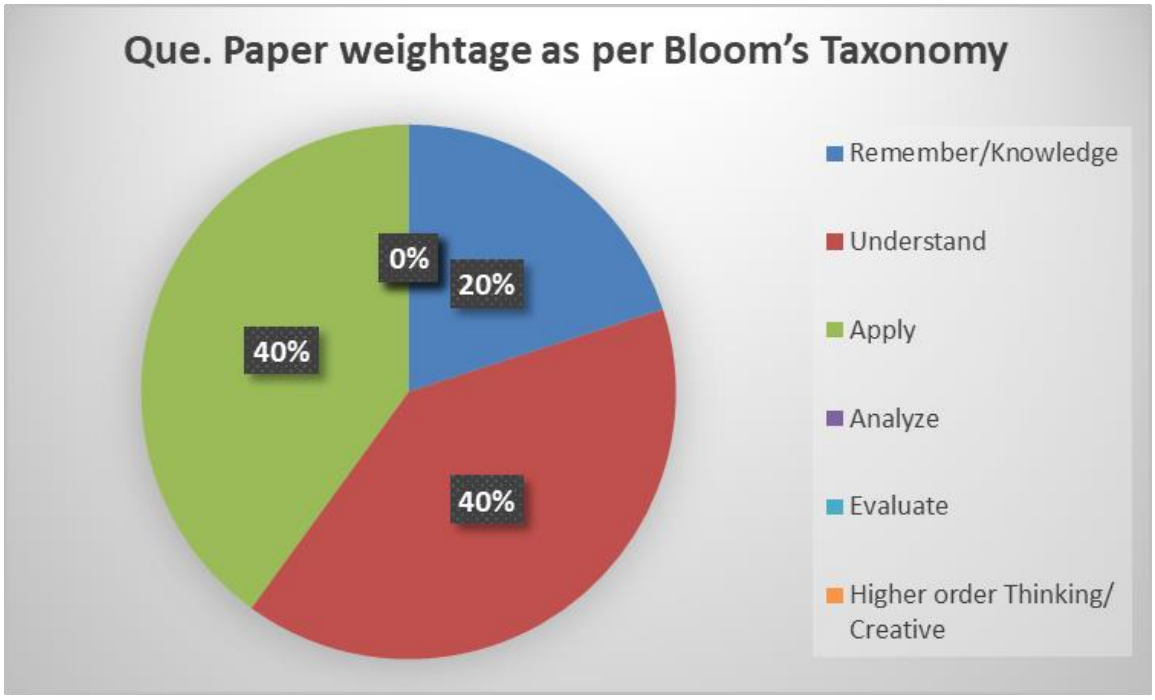
Bloom'S Taxonomy Report –

Sub: PS
Sem.3
Branch: CE, BIOINFO

Que. Paper weightage as per Bloom's Taxonomy

LEVEL	% of weightage	Question No.	Marks of Que.
Remember/Knowledge	20%	Q.1(a)	06
Understand	40%	Q.2(a), 2(b)	12
Apply	40%	Q.3(a), (b)	12
Analyze			
Evaluate			
Higher order Thinking/ Creative			

Chart/Graph of Bloom's Taxonomy



Course Outcome Wise Questions

Subject Code	**01CE0309**	Subject	**PROBABILITY AND STATISTICS**

CO No.	Course Outcome
CO1	Understand Basic probability concepts such as conditional probability, independence, and Bayes' theorem.
	1(A), 2(A), 3(B), 3(B-Or)
CO2	Apply probability distributions to solve problems involving random variables
CO3	Apply correlation and regression techniques to examine and interpret relationships between variables in engineering contexts.
	1(A), 2(B), 2(B-Or), 3(A), 3(A-Or)
CO4	Apply estimation techniques and construct confidence intervals in engineering problems
CO5	Analyze hypothesis tests and evaluate model fit using statistical methods.

Blooms Taxonomy	Question List
Remember / Knowledge	1(A)
Understand	2(A), 2(B), 2(B-Or)
Apply	3(A), 3(A-Or), 3(B), 3(B-Or)
Analyze	
Evaluate	
Higher order Thinking / Creative	